LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Quick Revision: How do Organisms Reproduce?

Why reproduction matters

- DNA copying passes on the body design. Small copying errors cause **variation**, which helps a species survive when the environment changes.
- Reproduction keeps populations stable in their niches.

Asexual reproduction (single parent)

Method	How	Example
Binary fission	Splits into 2 equal halves	Amoeba; Leishmania (definite plane)
Multiple fission	Splits into many cells at once, inside a cyst	Plasmodium (malaria parasite)
Fragmentation	Breaks into pieces, each grows into a new organism	Spirogyra
Regeneration	Body pieces grow into complete organisms (specialised cells)	Planaria, Hydra
Budding	Outgrowth (bud) grows and detaches	Hydra, yeast
Spore formation	Sporangia release spores	Rhizopus (bread mould)
Vegetative propagation	New plants from roots, stems or leaves	Bryophyllum (leaf notches), potato eyes, rose cuttings, grafting, layering

Vegetative propagation advantages: grows plants that don't make viable seeds (banana, orange, rose, jasmine); flowers and fruits come earlier; offspring keep the parent's good characters. **Tissue culture** grows many plants from a few cells.

Sexual reproduction in flowering plants

- Stamen (male): anther + filament. The anther makes **pollen**. Pistil (female): stigma, style, ovary with ovules (each ovule has an egg cell).
- **Pollination:** pollen reaches the stigma. Self-pollination is in the same flower; cross-pollination is to another plant (by wind, water or animals).
- Pollen tube grows through the style to the ovule. Male germ cell + female germ cell → **zygote** (fertilisation).
- After fertilisation: zygote → embryo; ovule → seed; ovary → fruit. Petals, sepals and stamens shrivel and fall.

Human reproduction

- **Puberty:** sex hormones (testosterone, oestrogen) bring changes in the body.
- **Male:** testes make sperm and testosterone (they lie outside the body because sperm need a lower temperature). Vas deferens carries sperm. Seminal vesicles and prostate add fluid.
- **Female:** ovaries release one egg a month and make oestrogen and progesterone. Fertilisation happens in the **oviduct**. The embryo implants in the **uterus**.
- **Placenta:** villi give a large surface area for glucose and O_2 to pass from mother to embryo, and wastes to pass back. Gestation is about 9 months.
- No fertilisation: the uterus lining breaks down and comes out as menstruation (about every month).

Contraception: barrier (condom, which also protects against STDs), hormonal (oral pills), copper-T/loop in the uterus, surgical (vasectomy / tubectomy).
STDs: gonorrhoea and syphilis (bacterial), warts and HIV-AIDS (viral).

Quick check

1. Name an organism that reproduces by multiple fission.

2. What do the ovule and the ovary become after fertilisation?

3. How does the placenta help the developing embryo?